



EMPLOYEE MANAGEMENT SYSTEM (SRS)

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Table of Contents

Section

1. Introduction

1.1 Purpose

1.2 Scope

1.3 Intended Audience

2. Overall Description

2.1 Users

3. Functional Requirements

3.1 User Authentication

3.2 Employee Management

3.3 Department Management

3.4 Attendance Management

3.5 Leave Management

3.6 Payroll Management

3.7 Notification Management

3.8 Dashboard & Reports

3.9 Profile Management

4. Non-Functional Requirements

4.1 Performance

4.2 Scalability

4.3 Security

4.4 Reliability

4.5 Availability

4.6 Maintainability

4.7 Usability

4.8 Compatibility

4.9 Data Integrity

5. Technical Specification

6. Acceptance Criteria

6.1 User Authentication

6.2 Employee Management

6.3 Department Management

6.4 Attendance Management

6.5 Leave Management

6.6 Payroll Management

6.7 Notification Management

6.8 Reports and Dashboard

6.9 Performance and Security

6.10 Overall System Acceptance

7. Conclusion

1. Introduction

The **Employee Management System (EMS)** is a web-based application to manage employee information and automate HR-related tasks within an organization. The system provides a centralized platform for managing employee records, attendance, leave requests, payroll, departments, and notifications. It improves efficiency, reduces manual work, and ensures secure access through rolebased authentication.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the Employee Management System. It serves as a reference for developers, testers, project managers, and stakeholders throughout the software development process.

1.2 Scope

The Employee Management System is designed to simplify employee and HR management by providing features such as employee record management, department management, attendance tracking, leave management, payroll management, notifications, and role-based access control. The system aims to improve productivity, data accuracy, and overall organizational efficiency.

1.3 Intended Audience

This document is intended for:

- Project Managers
- Software Developers
- Testers (QA Team)
- HR Personnel
- System Administrators
- Project Stakeholders

These users will use this document to understand the system requirements, functionalities, and overall project scope.

2. Overall Description

The Employee Management System (EMS) is a web-based application that centralizes employee and HR management activities within an organization. It enables administrators, HR personnel, managers, and employees to perform their respective tasks through a secure and user-friendly interface. The system is designed to automate routine operations such as employee record management, attendance tracking, leave management, payroll processing, and notifications, thereby improving efficiency and reducing manual work.

2.1 Users

The system supports the following user roles:

Admin

- Manages the entire system
- Creates departments and user accounts
- Monitors reports and system activities

HR

- Manages employee records
- Approves or rejects leave requests
- Processes payroll
- Sends notifications

Manager

- Manages employees within their department
- Reviews attendance
- Approves team leave requests

Employee

- Views personal profile
- Marks attendance
- Applies for leave
- Views salary details and notifications

3. Functional Requirements

3.1 User Authentication

The system shall provide secure authentication and authorization for all users.

Features:

- User Login
- User Logout
- Forgot Password
- Reset Password
- Role-Based Access Control (Admin, HR, Manager, Employee)

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3.2 Employee Management

The system shall allow authorized users to manage employee information.

Features:

- Add Employee
- Update Employee Details
- Delete Employee
- View Employee Details
- Search Employees
- Assign Employee to Department

3.3 Department Management

The system shall allow administrators to manage company departments.

Features:

- Create Department
- Update Department
- Delete Department
- Assign Manager
- View Department Employees

3.4 Attendance Management

The system shall record and manage employee attendance.

Features:

- Check-In
- Check-Out
- Manual Attendance
- Biometric Attendance (Optional)

View Daily Attendance

View Monthly Attendance

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- Calculate Working Hours

3.5 Leave Management

The system shall allow employees to request leave and enable managers/HR to review those requests.

Features:

- Apply for Leave
- View Leave History
- Approve Leave
- Reject Leave
- Leave Balance Management

3.6 Payroll Management

The system shall manage employee salary and payroll information.

Features:

- Generate Salary
- Calculate Net Salary
- Manage Allowances
- Manage Deductions
- View Salary History
- Download Payslip

3.7 Notification Management

The system shall allow authorized users to send and receive notifications.

Features:

- Send Company Announcements
- Leave Approval Notifications
- Salary Notifications
- Holiday Notifications

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- Email Notifications

3.8 Dashboard & Reports

The system shall provide dashboards and reports based on user roles.

Features:

- Employee Dashboard
- HR Dashboard
- Admin Dashboard
- Attendance Reports
- Leave Reports
- Payroll Reports
- Department Reports

3.9 Profile Management

The system shall allow employees to manage their personal information.

Features:

- View Profile
- Update Contact Information
- Change Password
- View Personal Attendance
- View Salary Details

4.Non-Functional Requirements

4.1 Performance

- The system should respond to user requests fast.
- The system should support multiple users simultaneously without significant performance degradation.

4.2 Scalability

- The system should support an increasing number of employees and departments.
- New modules and features should be added without affecting existing functionality.

4.3 Security

- User authentication shall be implemented using secure login mechanisms.
- Passwords shall be stored in encrypted form.
- Role-Based Access Control (RBAC) shall restrict unauthorized access.
- The system shall protect against SQL Injection and Cross-Site Scripting (XSS).

4.4 Reliability

- The system should provide accurate and consistent employee data.
- Database backups should be performed regularly.
- The system should recover gracefully from unexpected failures.

4.5 Availability

- The system should be available during working hours with minimal downtime.
- Scheduled maintenance should not affect stored data.

4.6 Maintainability

- The application should follow a modular architecture.
- The source code should be well documented.
- Bugs and updates should be easy to implement.

4.7 Usability

- The user interface should be simple and easy to navigate.
- Users should be able to perform tasks with minimal training.
- Error messages should be clear and informative.

4.8 Compatibility

- The system should support major web browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox.
- The application should run on Windows and Linux servers.

4.9 Data Integrity

- Employee information should remain accurate and consistent.
- Duplicate employee records should not be allowed.
- All important transactions should be validated before being saved.

5. Technical Specification

Frontend - Next JS

Backend - SpringBoot

Database - Postgre SQL

ORM - Hibernate

Build tool - Maven

Version control – Github & Git

Email Service – SMTP

IOT Biometric Device for attendance

6. Acceptance Criteria

The Employee Management System will be considered successfully implemented if it meets the following acceptance criteria:

6.1 User Authentication

- Users can log in using valid credentials.
- Invalid login attempts display appropriate error messages.
- Users can only access features based on their assigned roles (Admin, HR, Manager, Employee).

- Users can securely log out of the system.

6.2 Employee Management

- Admin and HR can add, update, and delete employee records.
- Employee details are stored accurately in the database.
- Employees can view only their own profile information.
- Employee records can be searched and filtered.

6.3 Department Management

- Admin can create, update, and delete departments.
- Employees are correctly assigned to departments.
- Department-wise employee lists are displayed accurately.

6.4 Attendance Management

- Employees can successfully check in and check out.
- Attendance records include date, check-in time, check-out time, and working hours.
- HR and Managers can view department attendance reports.
- Biometric attendance (if integrated) records attendance correctly.

6.5 Leave Management

- Employees can submit leave requests with a valid reason.
- Managers or HR can approve or reject leave requests.
- Leave status is updated correctly after approval or rejection.
- Employees can view their leave history.

6.6 Payroll Management

- HR can generate employee salaries.
- Salary calculations include allowances and deductions.
- Employees can view and download their salary details or payslips.
- Salary records are stored correctly for each payroll cycle.

6.7 Notification Management

- Admin and HR can send announcements and notifications.
- Employees receive notifications within the application.
- Email notifications are successfully delivered when email services are configured.

6.8 Reports and Dashboard

- Dashboards display information based on the user's role.
- Attendance, leave, employee, and payroll reports are generated correctly.
- Reports display accurate and up-to-date information.

6.9 Performance and Security

- The system responds to user requests within an acceptable time under normal load.
- Only authenticated users can access the system.
- Sensitive information such as passwords is securely encrypted.
- Unauthorized users cannot access restricted resources.

6.10 Overall System Acceptance

The Employee Management System shall be accepted if:

- All functional requirements are successfully implemented.
- All critical defects are resolved before deployment.
- System testing and user acceptance testing (UAT) are completed successfully.
- Data is stored and retrieved accurately.
- The application operates reliably without major errors during normal usage.

7. Conclusion

The Employee Management System aims to provide a centralized platform for managing employee records, attendance, leave requests, payroll, departments, and notifications. By automating HR processes and implementing role-based access control, the system will improve operational efficiency, reduce manual effort, and ensure secure and accurate management of employee information.